Autonomous Rapid Transit Feasibility in Toronto

Group 2: RALS

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Abstract:

Toronto and the Greater Toronto and Hamilton Area (GTHA) face a "missing middle" in their transit system, where medium-distance corridors (15–30 km) are underserved by slow local buses but lack density for subway investments. This leads to congestion, reduced accessibility, and economic losses, with TTC bus speeds declining 14% from 20 km/h in 2013 to 17.2 km/h in 2024 (City of Toronto, 2025), costing the TTC $40 million and the city $44.7 billion annually (Canadian Centre for Economic Analysis, 2024). Our research explores Autonomous Rapid Transit (ART), an emerging technology combining bus flexibility, rail capacity, and smoothness; and autonomous guidance on existing roads, as a cost-effective solution. Objectives include defining ART, analyzing trends, applications, challenges, and proposing a hypothesized implementation in Toronto. Methods involved utilizing peer-reviewed sources, case studies (e.g., Yibin, China; Perth, Australia, Auckland, New Zealand), and local data on congestion, debates, and emissions. Key findings highlight ART's potential for 170-500 passenger capacity per vehicle, speeds up to 70 km/h, and infrastructure costs at one-tenth of light rail, reducing GHG emissions, transportation accounts for 35% of Toronto's total (Feng et al. 2024; city of Toronto, 2025) and alleviating $44.7 billion in annual congestion costs (Canadian Centre for Economic Analysis, 2024). Recommendations include a pilot on Jane Street, with mitigations for risks like pavement rutting and geopolitical concerns. This positions ART as a transformative tool for equitable, sustainable urban mobility in Canada, aligning with net-zero 2040 goals and fostering economic returns of $7 per $1 invested (Miller, University of Toronto, 2025).

Introduction:

Autonomous Rapid Transit (ART), also known as a trackless tram, is an emerging battery-electric or hydrogen-powered transit technology that combines the high capacity and ride quality of light rail with the flexibility and lower infrastructure cost of a bus. Developed by the Chinese manufacturer CRRC, ART vehicles are bi-directional, 30–32 m long (3–5 modules), 2.65 m wide, and carry 300–500 passengers when coupled. They follow precise virtual tracks using multi-camera, LIDAR, GPS, and inertial navigation to shift pitch to achieve a smooth, train-like feel. ART achieves Level 4 autonomy in dedicated lanes or level 2 in shared traffic while retaining the ability for manual override (CRRC, 2024; Feng et al., 2024). With top speeds of 70 km/h, full traffic-signal priority, and zero tailpipe emissions, ART is positioned as a "missing-middle" solution between conventional buses and traditional steel-wheel light-rail transit (LRT).

The problem is framed using a Situation-Complication-Question (SCQ): The current situation is that The Greater Toronto and Hamilton Area (GTHA) continues to experience a persistent gap in medium-distance (15–30 km), medium-density transit corridors. Local bus speeds have fallen from 20 km/h in 2013 to 17.2 km/h in 2024, while bus on-time performance hovers in the low-to-mid 80% range and streetcar reliability remains below 70% for much of the year (City of Toronto, 2025; TTC, 2025). "Toronto currently operates only 33 km of bus lanes and 10 km of streetcar priority lanes…far behind peer cities such as New York (270+ km), Vancouver (72 km), and San Francisco (112 km)" (City of Toronto, 2025). This infrastructure deficit contributes to $44.7 billion in annual regional congestion costs and often weighs the most on disproportionately affected communities. Those who rely on surface transit and cannot afford and maintain a private car (Canadian Centre for Economic Analysis, 2024; Ipsos, 2024).

Complication: Traditional rail-based solutions have become increasingly expensive and delayed in English-speaking states. Toronto's 19 km Eglinton Crosstown LRT, originally budgeted at $8.2 billion in 2010, is now estimated to exceed $13 billion with opening repeatedly postponed (Casey, 2025). The 11 km Finch West LRT, delivered under a public-private partnership (P3), opened in December 2025 at a final cost of approximately $3.7 billion-$1.2 billion (48%) above the original $2.5 billion estimate (DCN-JOC News Services, 2025). Even after opening, early

operations have been criticised for end-to-end travel times of 46–55 minutes (versus the projected 34 minutes) due to conservative usage, traffic-signal priority, and lowered speed during the introductory phase (Lang, 2025). Additionally, the TTC lowered 25% fleet-wide GHG emission reductions since 2017 by adding in electric buses. However, The recent launch of Finch West LRT and Electric buses still get trapped in traffic, adds to congestion, and wastes energy, time, and transit resources. Without a new solution, the GTHA will be unable to achieve its transportation, economic, and climate goals with current technologies, leaving thousands of commuters underserved and worsening the region's mobility crisis. As for light and heavy rail

Question: How can emerging technologies like ART fill this gap to achieve faster, equitable transit while meeting climate goals (e.g., 35% of GHG from transportation; City of Toronto, 2025)?

Research Body:

1. Technology Overview

Autonomous Rapid Transit (ART) represents an aggregate of multiple emerging technologies designed to address the limitations of conventional mass transit systems. As documented by Feng et al. (2024), ART vehicles are bi-directional, multi-articulated buses measuring 30-32 meters in length and 2.65 meters in width, with a capacity of 170-300 passengers in standard configuration, expandable to 500 passengers when multiple units are coupled. The vehicle uses multi-sensor guidance, including optical cameras, LIDAR, high-precision differential GPS, and inertial measurement units to achieve Level 4 autonomy in dedicated lanes while maintaining driver manual override capability (CRRC, 2024; HandWiki, n.d.).

An ART either uses lithium-titanate batteries with 600 kWh capacity providing 40 km range per charge, or hydrogen fuel cells offering extended range up to 390 km (CRRC, n.d.). The lithium-titanate chemistry specifically addresses Toronto's climate challenges, maintaining performance at temperatures in mild Chinese winters. The charging infrastructure requirements are spread into three parts: 30-second opportunity charging at stations providing 3-5 km range, 10-minute terminal charging delivering 25 km range, and at the end of the stop for full battery restoration (Feng et al., 2024). To support these operations, the station service system is designed with integrated power supply and safety monitoring modules to ensure seamless vehicle-platform interaction (Zou et al., 2020). Low infrastructure costs and no rail construction enables

deployment at approximately one-tenth the infrastructure cost of conventional LRT systems. Newman (2020) classifies the technology as a "disruptive innovation," noting that Trackless Trams retain the ride quality of light rail but can be implemented for approximately $5 million per km compared to $50–80 million for traditional rail projects. or $10-40 million per kilometer versus $1.6 Billion for LRT (Eglinton LRT) (School of Cities, University of Toronto, 2025).

Current Trends

The global transit construction landscape reveals a stark disparity in costs and timelines, with English-speaking nations experiencing 5-10 times higher costs than comparable projects in Europe and Asia (School of Cities, University of Toronto, 2025). Toronto exemplifies this trend with the Eglinton Crosstown LRT, originally budgeted at $8.2 billion in 2010, now exceeding $13 billion with repeated delays (Casey, 2025; Thayaparan, 2025). The recently opened Finch West LRT demonstrates similar challenges, with final costs of $3.7 billion representing a 48% increase over initial estimates and early operations achieving only 46-55 minute end-to-end travel times versus the projected 34 minutes (DCN-JOC News Services, 2025; Lang, 2025).

In China, ART deployment has accelerated significantly since 2017, with China operating over 140 km of ART routes across five cities, carrying more than 35 million passengers and traveling 25 million kilometers while saving 24,500 tonnes of CO_2 compared to diesel buses (CNBC, 2023). The Yibin deployment, operational since 2021, demonstrates mature system performance with 30,000 daily riders, 95% on-time performance, and 85% passenger satisfaction while achieving 40% lower operational costs than conventional buses (Jia et al., 2025).

In Australia, Perth's 2023 Scarborough corridor trials yielded 94% of riders rating ART ride quality as "good to excellent," with 76% of residents supporting adoption for future public transit (Public Transport Association of Canberra, 2021). However, political priorities favoring traditional heavy rail prevented funding allocation despite positive trial results, mirroring Toronto's Scarborough subway debate, where evidence-based solutions were overridden by political symbolism (CBC, 2019; University of Toronto Transportation Research Institute, 2015).

In North America, CRRC's own two U.S. facilities demonstrate both operational viability and geopolitical hurdles. The Springfield, Massachusetts plant (CRRC MA) focuses on MBTA

Red/Orange Line cars for Boston and a smaller order for Los Angeles Metro, employing over 400 workers despite recent U.S. The Chicago, Illinois, facility exclusively builds and replaces half of the CTA's 'L' trains under a $1.3 billion contract, remaining active through at least 2028 with no reported disruptions (Rail Security Alliance, 2020). CRRC's bids were 20–50 % cheaper than competitors like Bombardier and Kawasaki. This limited but stable U.S. footprint makes ART's potential for localized manufacturing partnerships in Canada, mitigating similar procurement risks.

Applications

ART's primary application addresses the "missing middle" gap in metropolitan transit networks corridors of 15-30 km that exceed efficient local bus service range but lack density for subway investment (Miller et al., 2025). In the GTHA context, where 480,000 additional commuters are projected by 2031, representing a 41% increase from 2016 levels, ART provides critical capacity expansion without the decade-long construction timelines of conventional rail (Canadian Centre for Economic Analysis, 2023).

The technology excels in arterial corridor transformation, converting congested mixed-traffic roadways into dedicated rapid transit routes. Analysis of Toronto's Jane Street corridor, carrying 48,000-55,000 daily riders on routes 35/935, demonstrates immediate application potential where current bus speeds average 17.2 km/h but ART could achieve 40-50 km/h with dedicated lanes and signal priority (City of Toronto, 2025; Toronto Transportation Services, 2025). This speed improvement translates to 20-30 minute reductions in commute times for corridor residents, with particular benefit to equity-deserving communities where trips lasting 60+ minutes are twice as common as trips under 30 minutes (Ipsos, 2024).

Transit-oriented development (TOD) applications leverage ART's station infrastructure to catalyze urban intensification. Research indicates property value uplifts of 5-20% for residential properties and up to 12% for transit-accessible areas within 600 meters of rapid transit stations (Miller et al., 2025). Newman (2020) argues that this technology allows planners to "mainstream" climate-responsive design, unlocking urban regeneration potential in corridors that previously lacked the density to justify expensive rail infrastructure. Early designs of ART

station service systems further enable efficient passenger handling and integration (ZOU et al., 2020).

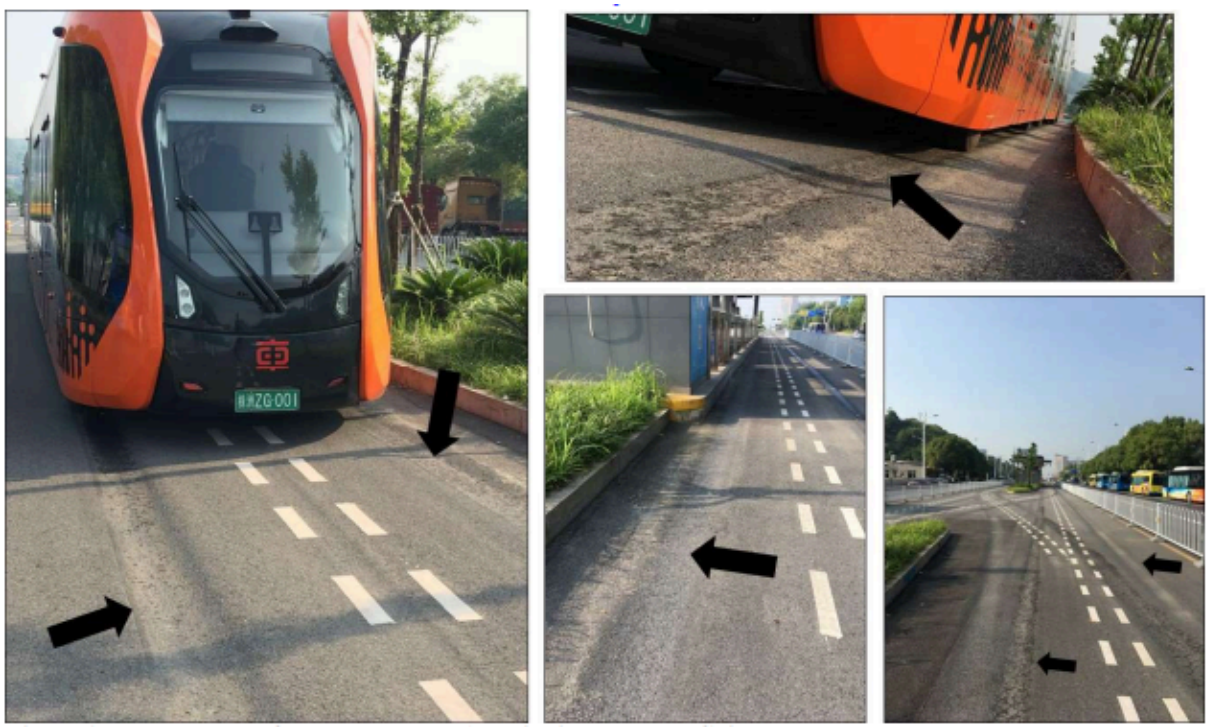
Integration with existing transit networks positions ART as a complementary layer rather than a replacement technology. Local buses function as feeder services to ART corridors, while ART extends the reach of subway terminals and provides cross-suburban connections, avoiding downtown routing through Union Station, currently a capacity bottleneck for GO Transit's radial network (Metrolinx, n.d.). This multi-modal integration addresses the structural limitation where 70-90% of GTHA commutes utilize private vehicles due to inadequate suburban transit connectivity (Canadian Centre for Economic Analysis, 2024).

Challenges & Limitations

Technical challenges include winter operational reliability, given limited cold-weather testing data, with sensor occlusion from snow and ice potentially compromising the optical guidance systems (Data Insights Market, n.d.). CRRC claims robust winter performance and lithium-titanate batteries offer superior cold-weather characteristics (Feng et al., 2024). CRRC has received some International Certifications, However independent Canadian testing will be done on trial period. These tests will include 40 km standard battery range, realistically 30 km accounting for heating and winter operations.

Critical to ART's infrastructure compatibility is its ability to operate on existing road surfaces without the rail installation required for traditional light rail transit (LRT). However, peer-reviewed independent studies have found inconsistent results. According to the United Nations Centre for Regional Development assessment, over three years of Chinese operational data show no significant pavement degradation from ART operations (UNCRD, n.d). But the Public Transport Research Group at Monash University, Australia, in 2021 found road rutting “varies significantly with both the general type of road construction as well as the quality of the

‘subgrade’ underneath the road pavement.”



The figure above shows significant rutting on ART deployment. CRRC commented that this implementation was rushed, also admitting that “That future implementation would include works to strengthen road pavements." Infrastructure concerns persist despite manufacturer claims of minimal pavement impact. The Public Transport Association of Canberra (2021) documented pavement rutting concerns in early deployments while planning in Perth, while Auckland's technical assessment concluded ART's axle loads would require expensive road reinforcement, eliminating projected cost savings. Toronto's testing would include potentially reinforced pavement specifications on account of weight and potential winter deterioration.

Institutional barriers represent significant implementation challenges, including unclear regulatory pathways for autonomous vehicle operation, with federal and provincial frameworks lagging technological capability (Ubaidillah et al., 2024). Labor relations require careful management given potential displacement of bus operators, though U.S. CRRC facilities demonstrate successful integration of unionized workforces in manufacturing and operations (CRRC, n.d.). Inter-municipal coordination between Toronto, York Region, Peel Region, and Metrolinx introduces complexity, given overlapping jurisdictions and competing priorities.

Geopolitical risks cannot be ignored, given CRRC's Chinese state ownership and precedents, including the U.S. Congress's 2020 CRRC ban and Canada's Huawei 5G exclusion (CBC, 2022). While mitigation strategies include domestic manufacturing partnerships leveraging Canada's existing P3 infrastructure delivery model used for Eglinton Crosstown and Ontario Line projects, technology transfer and cybersecurity concerns require a comprehensive risk assessment and potential supply chain diversification.

Social acceptance barriers identified by Ubaidillah et al. (2024) indicate that in car-centric cultures like GTHA suburbs, normative factors outweigh individual attitudes in determining ART adoption likelihood. This necessitates intensive community engagement emphasizing peer usage, environmental responsibility, and modern identity construction, a 2-3 year social normalization campaign paralleling infrastructure development.

Hypothesized Solution

RALS proposes a phased ART implementation framework beginning with a 12-14 km pilot corridor on Jane Street from Eglinton Avenue West to Pioneer Village Station, leveraging the corridor's existing political momentum, utilizing newly included “RapidTO” that has priority TSP and dedicated bus lanes. (City of Toronto, 2025). This first deployment would demonstrate ART's technical viability while addressing the most acute transit inadequacy in Toronto's network, where routes 35/935 Jane consistently rank among the TTC's highest-ridership yet poorest-performing services.

The implementation strategy employs a three-tier risk mitigation approach. First, technical validation through 18-month pre-deployment testing, including winter trials, cybersecurity audits, and pavement monitoring, addresses operational uncertainties. Second, institutional alignment via collaborative agreements with labor unions, guaranteeing no involuntary layoffs and retraining programs, following successful models from U.S. CRRC facilities employing unionized workers (Feng et al., 2024). Third, financial de-risking through mixed procurement options including CRRC manufacturing in Canadian facilities staffed by Canadian workers, similar to established P3 models, with parallel exploration of licensing agreements with Bombardier or European suppliers like Alstom or Siemens.

Infrastructure deployment would occur in parallel tracks: physical infrastructure (dedicated lanes, stations, and service systems [Zou et al., 2020]) proceeding through conventional procurement, estimated at $120-350 million for the Jane corridor, while vehicle acquisition and system integration follow a performance-based contract structure linking payments to operational metrics. This approach limits public sector exposure while incentivizing private partner performance, proven effective in recent Metrolinx projects (Infrastructure Ontario, n.d.).

Community engagement employs evidence-based social marketing emphasizing three key messages validated by Ubaidillah et al. (2024): environmental responsibility (35% GHG reduction from transportation), and future-forward identity ("Toronto leads in sustainable mobility"). Early engagement with Jane-Finch and Black Creek communities ensures equity-deserving populations shape service design rather than having solutions imposed.

The success metrics framework establishes clear, measurable targets: achieving 40-50 km/h average speeds (versus current 17.2 km/h), >90% on-time performance (versus current 76-77%), 22,000-28,000 daily ridership within two years, and 15-20% mode shift from private vehicles. Economic indicators include 5-15% property value uplift within 600m of stations and $50-80 million annual congestion cost savings through improved corridor flow (Miller et al., 2025; Canadian Centre for Economic Analysis, 2024).

Future Outlook

The long-term adoption trajectory for ART in the GTHA depends on successful demonstration of technical reliability, public acceptance, and political sustainability beyond individual election cycles. Global market analysis projects compound annual growth rates of 15-20% in the autonomous transit sector through 2033, driven by urbanization pressures and climate imperatives (Data Insights Market, n.d.). For Toronto specifically, achieving 2040 net-zero targets requires reducing transportation emissions by 35%, necessitating a modal shift from private vehicles to high-capacity electric transit, a transformation ART can accelerate given its 2-3 year deployment timeline versus 10-15 years for conventional rail (City of Toronto, n.d.).

Technological evolution toward Level 5 full autonomy could eliminate operator costs entirely, reducing operational expenses by 30-40% based on current TTC labor costs of approximately $100,000 per operator annually, including benefits (Feng et al., 2024). Integration with smart city infrastructure, including real-time traffic management, predictive maintenance, and demand-responsive routing could further enhance system efficiency. The convergence of ART with mobility-as-a-service (MaaS) platforms positions the technology as a backbone for integrated multi-modal journey planning, particularly as younger demographics demonstrate decreasing car ownership rates.

Canadian readiness levels vary significantly across technological, institutional, and social dimensions. Technologically, Canada's strong engineering sector and cold-weather expertise position the nation to become a global leader in winter-capable autonomous transit, potentially developing exportable intellectual property. However, institutional readiness lags with fragmented transit governance, risk-averse procurement practices, and limited regulatory frameworks for autonomous vehicles. The School of Cities report (2025) identifies systemic factors driving Canada's transit cost premiums, suggesting institutional reform must accompany technological adoption.

The proposed 10-year vision of 100-150 km of integrated ART network across the GTHA would fundamentally restructure regional mobility, creating a medium-capacity rapid transit layer complementing existing subway, LRT, and GO networks. Phase 2 corridors, including Dufferin Street (11-13 km), Steeles Avenue West (15-20 km), and Sheppard Avenue East (12-15 km), would demonstrate inter-municipal coordination capacity while serving diverse demographic and geographic contexts. Phase 3 expansion to Finch Avenue East, Lawrence Avenue, Wilson Avenue, and Eglinton East would complete a grid network enabling true suburb-to-suburb connectivity without downtown routing.

Innovation opportunities extend beyond transportation to urban development patterns. ART's lower infrastructure costs enable transit-oriented development in previously unviable locations, potentially reversing decades of car-dependent sprawl. The land use efficiency gains reducing parking requirements from 4-10 spaces per person to less than one per household, freeing

substantial urban land for housing, parks, and economic development (Feng et al., 2024). This transformation aligns with provincial housing targets requiring 1.5 million new homes by 2031, with transit-accessible development commanding premium values and reducing per-unit infrastructure costs.

Research priorities should focus on addressing identified knowledge gaps: comprehensive winter performance validation, Canadian-specific pavement impact studies, cybersecurity frameworks for transit systems, and social acceptance factors in multicultural contexts. Academic partnerships between the University of Toronto's Transportation Research Institute, Durham College's emerging technology programs, and international research networks could position the GTHA as a living laboratory for next-generation transit innovation. The economic returns of $7 per dollar invested in transit operations justify sustained public and private research investment (Miller et al., 2025).

Conclusion:

The Greater Toronto and Hamilton Area stands at a critical inflection point where traditional transit planning approaches have proven inadequate to address the "missing middle" challenge that leaves 15-30 km corridors underserved, perpetuates $44.7 billion in annual congestion costs, and undermines climate commitments requiring 35% transportation emission reductions by 2040 (Canadian Centre for Economic Analysis, 2024; City of Toronto, n.d.). The decline of TTC bus speeds by 14% to 17.2 km/h since 2013, combined with reliability low-to-mid 80% range and streetcar reliability remains below 70% for much of the year, demonstrates that incremental improvements to existing systems cannot meet projected demand growth of 480,000 additional commuters by 2031 (City of Toronto, 2025; Canadian Centre for Economic Analysis, 2023).

Autonomous Rapid Transit emerges as a transformative solution uniquely suited to Canadian urban contexts, offering rapid transit quality at one-tenth the infrastructure cost of LRT and 2-3 year deployment timelines versus the decade-long construction periods plaguing projects like Eglinton Crosstown (Feng et al., 2024; School of Cities, University of Toronto, 2025). The technology's proven operational performance in Yibin, 30,000 daily riders, 95% on-time performance, 40% operational cost savings versus buses, combined with successful trials in Perth

demonstrating 94% passenger satisfaction, validates ART's technical maturity and public acceptance potential (Jia et al., 2025; Public Transport Association of Canberra, 2021).

Our proposed implementation framework, beginning with the Jane Street corridor serving 48,000-55,000 existing daily riders, addresses technical uncertainties through rigorous testing, institutional barriers via collaborative labor agreements, and financial risks through mixed procurement strategies. The projected benefits of 20-30 minute commute reductions, 5-15% property value uplifts, $50-80 million annual congestion savings deliver economic returns of $7 per dollar invested while advancing equity by serving transit-dependent communities currently facing 60+ minute commutes (Miller et al., 2025; Ipsos, 2024).

The path forward requires political courage to embrace innovation over conventional solutions that have consistently exceeded budgets and timelines. ART is not merely a new vehicle technology but a paradigm shift in how cities approach mobility infrastructure, flexible, scalable, and achievable within political cycles. As Toronto faces the dual imperatives of explosive growth and climate crisis, the question is not whether to adopt emerging transit technologies, but whether we can afford the economic, social, and environmental costs of maintaining the status quo. The evidence overwhelmingly supports immediate action: ART represents infrastructure rethought for the challenges of the 21st century city.

AI Usage:

This research paper was developed with assistance from AI tools. AI was utilized for formatting assistance. Idea structure and sentence grammar corrections to improve the written content. All content, including research, analysis, arguments, and conclusions, was made entirely by the RALS team.

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Responsibility Matrix

Name	Roles/Contributions	Date
Riyaan Hassan	Framework & Designer: Created the document layout and starting point. Established the initial structure and organized the document flow. Wrote Ai usage as described by the teams overall Ai assistance.	Dec 12, 2025
Abdul Hadi	Editor and Writer (Problem, Conclusion, Applications & Trends): Authored the "Current Trends" section. Analyzed GTHA congestion data, financial statistics (Eglinton/Finch costs), and global case studies (China/Australia). then conducted the final review for flow and grammar. Managed APA citations, the Reference list, and final formatting/editing.	Dec 12, 2025
Lucas Delvoie	Researcher & Writer Wrote the Abstract, Future outlook and Introduction. Researched ART technical specifications (CRRC data, sensors, battery types, charging, etc) and compiled the "Technology Overview" section.	Dec 12, 2025
Subeet Singh	Critical Analyst (Challenges, Hypothesized Solution, & Conclusion): Wrote "Challenges & Limitations" (infrastructure/geopolitical risks).	Dec 12, 2025